Outage Investigation Report — OIR-2025-010

Field	Value
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Region	CAISO
Site	CAISO-WIND-026
Asset	WIN-10130 (Siemens Gamesa, wind_turbine)
Outage IDs	OUT-2025-0556, 0566, 0558, 0559 (+ 10 additional, FY-summary below)
Classification	Multi-event roll-up — overheating + vibration_excess
Distribution	CAISO Operations, NW-PowerPool Reliability Engineering, OEM Liaison, Vendor Management, Executive Operations Committee

1. Outage Summary

This report is a **multi-event roll-up** for WIN-10130 at CAISO-WIND-026 covering all 14 FY2025 forced outages. Cumulative lost generation **20,494 MWh** — the highest single-asset lost generation in the Cascadia fleet for FY2025. Estimated revenue loss $1,229,665.

WIN-10130 is a Siemens Gamesa unit identical in model and build year to the four problem turbines at NW-PowerPool-WIND-024 (WIN-10011, -10131, -10054, and one other). The cumulative failure pattern is the subject of RCA-2025-003.

Highest-impact events on WIN-10130 in FY2025
2025-03-09 — overheating — OUT-2025-0556 — 213 h — 2,883.3 MWh
2025-05-16 — vibration_excess — OUT-2025-0566 — 211 h — 2,856.2 MWh

Highest-impact events on WIN-10130 in FY2025
2025-06-21 — vibration_excess — OUT-2025-0558 — 202 h — 2,734.4 MWh
2025-10-10 — vibration_excess — OUT-2025-0559 — 183 h — 2,477.2 MWh
10 additional smaller events — cumulative 12,043 MWh

2. Timeline of Events (Yearly Overview)

Window	Event class	Notes
2025-01-20	First FY25 event — overheating	Restart on 2025-01-26 after CPS repair. CPS dispatched from Bakersfield depot.
2025-03-09 → 2025-03-18	Overheating, 213 h	Gearbox oil change, heat exchanger inspection. Apex via EPA.
2025-05-16 → 2025-05-25	Vibration_excess, 211 h	Siemens Gamesa OEM activated under Rev. 4.2 manual. OEM recommended full gearbox replacement at $1.2M.
2025-06-21 → 2025-06-29	Vibration_excess, 202 h	OEM gearbox replacement quote escalated to executive committee. Internal engineering counter-proposes refurbishment plus Cascadia threshold override (basis for MOC-2025-022).
2025-07 to 2025-09	5 smaller events	Heartland Reliability Partners dispatched on 3 of 5; Apex on 2. Mixed outcomes.
2025-10-10 → 2025-10-18	Vibration_excess, 183 h	Vibration RMS at 7.1 mm/s — above MS-2025-03 limit 6.5, below OEM threshold 10.0. Same procedural conflict as NW-PowerPool.
2025-12-07	Final FY25 event	Pitch-system fault, 4-day outage.

3. Telemetry Observations (Aggregate)

- Vibration RMS on main shaft varied between 5.2 and 7.4 mm/s across the year. The asset operated above MS-2025-03 alert threshold (6.5) for 178 cumulative hours.
- Gearbox sump oil temperature averaged 76 °C under normal load — 4 °C above commissioning baseline. Heat exchanger fouling suspected but inspections have closed without findings.
- Iron particulate trend: rose from 6 ppm in January to 14 ppm in October. Above limit (12) since August.

- Per RCA-2025-003 analysis: this asset's degradation pattern is consistent with end-of-economic-life behavior. Twelve sister-units at CAISO-WIND-026 and NW-PowerPool-WIND-024 show similar trend slope.

4. Restoration Activities (FY25 Aggregate)

Vendor	Events handled	Total invoiced (USD)
Cascadia Power Services (CPS)	4	$312,000
Apex Emergency Grid Solutions	6	$1,910,000
Apex / Siemens Gamesa subcontract (1.8×)	3 (subset of Apex above)	(included)
Heartland Reliability Partners	3	$284,000
Siemens Gamesa OEM direct	1 (warranty review)	$112,000
Total	14	**$2,618,000**

Average per-event Apex invoice on this asset: $318K. CPS equivalent scope estimates averaged $94K — consistent with the 4× arbitrage discussed in EMR-2025-002 and RCA-2025-001.

5. Preliminary Root Cause

This asset exhibits **end-of-economic-life degradation** rather than a single discrete failure mode. The aggregate vibration, oil-particulate, and temperature trends indicate cumulative wear on the gearbox and main bearing assembly. Repair-cycle replacement has been progressively less effective; the gap between repair-quality and pre-failure state is narrowing.

Contributing factors:
1. Siemens Gamesa OEM Manual Rev. 4.2 specifies replacement at vibration RMS 6.0 mm/s on the high-speed shaft. Standard MS-2025-03 sets the threshold at 8.0 mm/s. The asset has spent FY2025 operating between these two values; the OEM has issued replacement quotes; Cascadia engineering has rejected them. **MOC-2025-022 is the pending instrument to formally adopt the Cascadia threshold or to formally adopt the OEM threshold.** As of report date (2026-01-09) MOC-2025-022 has been pending engineering review for 10 months.
2. The asset is operating under a 4-month vendor mix (CPS-then-Apex-then-Heartland) that has prevented coherent service history. This is a vendor governance concern; refer to RCA-2025-003.

3. The asset is one of 12 identical sister-units. A coordinated FY26 replacement or major overhaul plan should be the recommendation rather than continued reactive repair.

6. Recommendations

1. **Approve MOC-2025-022.** Either adopt the OEM threshold and budget the $1.2M gearbox replacement, or formally adopt the Cascadia MS-2025-03 threshold and discontinue OEM quote review. The current limbo is the most expensive option.
2. Initiate the FY26 capital planning workstream for the 12-unit sister-fleet (WIN-10130, -10131, -10011, -10054, plus 8 others). See RAM-2025-007 (in preparation).
3. Settle vendor mix at CAISO-WIND-026. Recommend single-prime arrangement under CPS or Heartland for the next 12 months to establish coherent service history. Refer to VRA-2025-006.
4. Update OIR template to require explicit MS-2025-03 vs OEM threshold comparison in §3 telemetry section.
5. Cross-reference RCA-2025-003 (WIN-10130 chronic failure pattern), OIR-2025-002 (WIN-10131 August event, sister-unit), OIR-2025-009 (WIN-10011 December event, sister-unit).

Signatures

Role	Name	Date
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Distribution: CAISO Operations · NW-PowerPool Reliability Engineering · OEM Liaison · Vendor Management · Executive Operations Committee · CFO Office

References: OIR-2025-002 · OIR-2025-005 · OIR-2025-009 · RCA-2025-001 · RCA-2025-003 · MOC-2025-022 · RAM-2025-007 (draft) · VRA-2025-006 · EMR-2025-002 · Siemens Gamesa OEM Manual Rev. 4.2 · Standard MS-2025-03